

TASK 3 — Body Segmentation Methods

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Project Folder: Raritone

1. Introduction

Body segmentation is a process of separating the human body from the background in images or videos. It is widely used in computer vision applications.

2. Working Principle

Segmentation works by classifying each pixel as either belonging to the human body or the background.

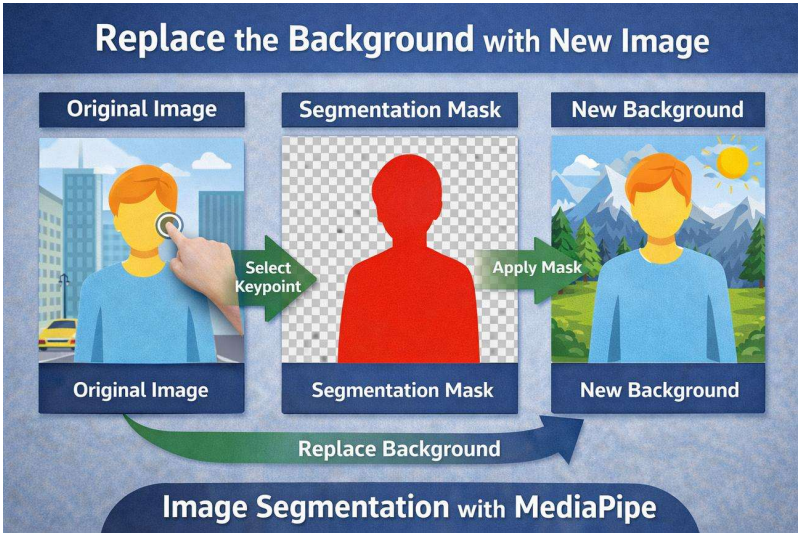
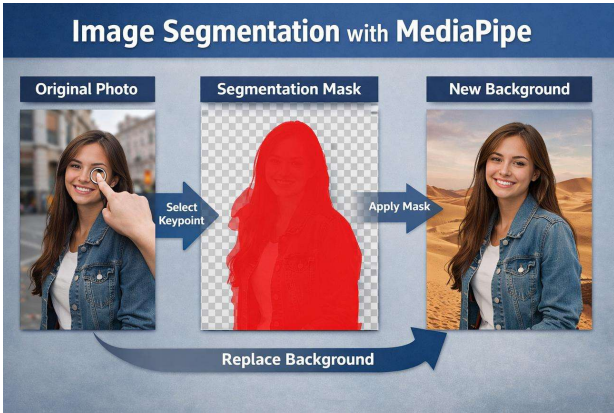
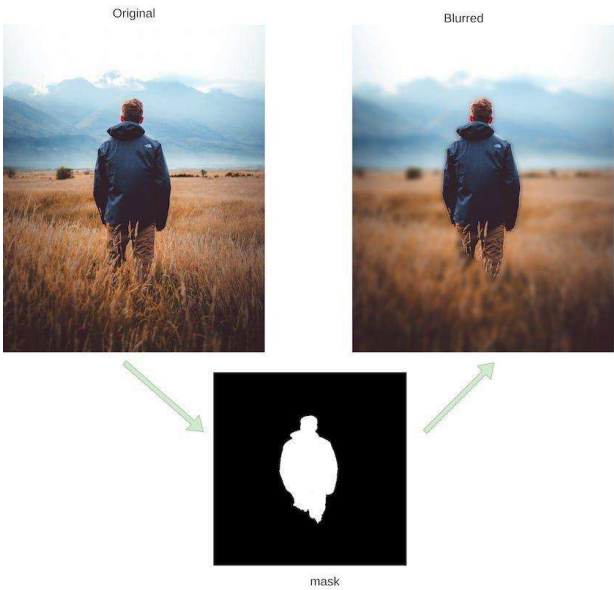
3. Implementation

In this project, MediaPipe's segmentation mask was used.

Steps:

- Capture video frame
- Generate segmentation mask
- Apply threshold
- Modify background

4. Output



5. Results

- **Background successfully blurred**
- **Clear separation of subject**
- **Enhanced visual output**

6. Applications

- **Virtual try-on**
- **Video conferencing**
- **AR filters**

7. Limitations

- **Sensitive to lighting**
- **Errors in complex backgrounds**
- **Performance depends on model quality**

8. Conclusion

Body segmentation is an essential technique that enhances visual clarity and enables advanced AI applications.